

S.No. : 20

BCS 2604

No. of Printed Pages : 06

Following Paper ID and Roll No. to be filled in your Answer Book.

PAPER ID : 23220

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No.

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B. Tech. Examination 2021-22

(Even Semester)

COMPILER DESIGN

Time : Three Hours]

[Maximum Marks : 60

Note :- Attempt all questions.

SECTION – A

1. Attempt all parts of the following : $8 \times 1 = 8$
 - (a) What is role of code optimization in compiler?
 - (b) Define the term bootstrapping.
 - (c) Mention the rule of eliminate left recursion.
 - (d) Short notes on evaluation order of semantics action in SDD.
 - (e) Define the forms of intermediate code.
 - (f) Define the role of lexical analyzer in compiler.

[P. T. O.

- (g) DAG for expression $(a * b) + (a * b)$ using mknnode and mkleaf function.
- (h) Write down the short notes on sub common expression elimination.

SECTION – B

2. Attempt any two parts of the following : $2 \times 6 = 12$

- (a) Construct the syntax tree using syntax directed definition for the expression $x * y - 5 + z$ with the help of grammar G as follows :

G :

$E \rightarrow E + T / E - T / E * T / T$

$T \rightarrow \text{id} / \text{digit}$

- (b) Check whether the given grammar is CRL (1) and LALR (1) or not? Grammar G as follows :

G :

$s \rightarrow Aa$

$s \rightarrow bAc$

$s \rightarrow Bc$

$s \rightarrow bBa$

$A \rightarrow d$

$B \rightarrow d$

- (c) Parse the given input string $id + id * id$ using LL (1) parser for the given grammar G as :

G :

$E \rightarrow E + T / T$

$T \rightarrow T * F / F$

$F \rightarrow id$

- (d) Describe the role of optimization techniques.

SECTION – C

Note :- Attempt all questions from this section.

$10 \times 4 = 40$

3. Attempt any two parts of the following :

- (a) Generate the intermediate code for Boolean expression as follows :

$A > B \text{ AND } B < C \text{ OR } G < H$

with proper SDD or useful production rule of general grammar.

- (b) Write down the SDD for while-do statement and generate three address code for given expression :

[P. T. O.]


```
while (A < C and B > D) do
{
    if A = 1
        then A = A + 1
    Else B = B + 1
}
```

- (c) Write the quadruples, triple, and indirect triple intermediate code for the following expression :

$$(x + y) * (y + z) + (x + y + z)$$

4. Attempt any two parts of the following :

- (a) Calculate First and Follow function for the given grammar G as :

```
G :
A → AcB / cC / C
B → bB / d
C → caB / BbB / B
```

- (b) Construct operator precedence parser and then parse the following string (a, (a, a)). Grammar G as :

G :

$S \rightarrow (L) / a$

$L \rightarrow L, S / S$

- (c) Check whether the given grammar is CLR (1) or not? Grammar G as :

$$S \rightarrow CC, C \rightarrow a C/d$$

5. Attempt any two parts of the following :

- (a) Write down the differences between the synthesized and inherited attribute with suitable examples.
- (b) Construct syntax tree for statement

$$x * y - 5 + z$$

given grammar G using `mk leaf ( )`, `mknode ( )` functions :

G :

$E \rightarrow E + T / E - T / E * T$

$T \rightarrow \text{num} / \text{id}$

- (c) Explain the role of optimization in compiler and define all types of loop optimization techniques used for basic blocks.

[P. T. O.]

6. Attempt any two parts of the following :

- (a) Explain the phases of compiler and implement all the phases for given statement :

$$x := a + b - c * 9$$

- (b) Generate code for the following statement using simple code generation algorithms :

$$x = a + \frac{(b + c / d * e)}{(f * g)}$$

- (c) Write short notes on the following :

- (i) Optimization of basic blocks
- (ii) Loop unrolling and loop fusion
